Lucas Tolman & Dhruv Laungani
DS4420 - Dr. Gerber

DETECTING SONG “BREAKOUT” ON SPOTIFY

Predicting Charts Success from Initial Stream Trajectories Utilizing AR/MA and Bayes ML

Northeastern University

PURPOSE

Can we predict whether a song will reach the Top 10 after just 2-3 weeks on Spotify charts? We build a two-stage pipeline:

1. Fit **AR/MA** models to early streaming data to extract momentum features
2. Combine trajectory features with audio attributes in a **Bayesian logistic regression** to output P(Top 10)

BACKGROUND

Prior work either uses a song's full streaming history (only available in hindsight) or static audio features alone (ignores early momentum).

We chose two Kaggle datasets: Spotify Charts (26.2M daily entries, 2017-2021) and Spotify Tracks (114K songs with audio features). The cleaned dataset contains 3,255 songs with 21-day stream windows.

METHODS

AR/MA Time Series using **Python**

- Extract first 21 days of daily streams per song
- Apply first-order differencing to remove trend
- ACF/PACF analysis on example songs confirms ARMA(1,1)
- Fit AR(1) **manually** via OLS
- Fit MA(1) via statsmodels

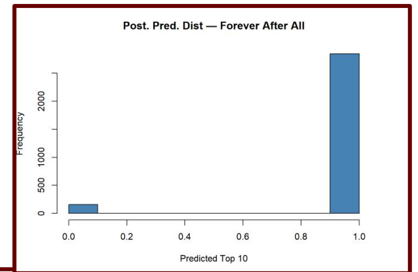
Bayesian Logistic Regression using **R**

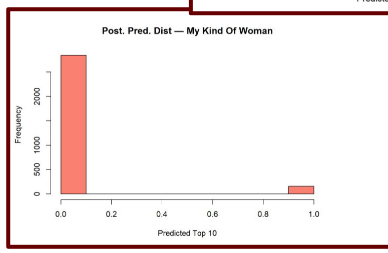
- Predictors: trajectory features + audio attributes, standardized
- Priors: N(0,2) on coefficients, Student-t(4,0,5) on intercept
- MCMC: 4 chains, 2000 iterations, 500 warmup

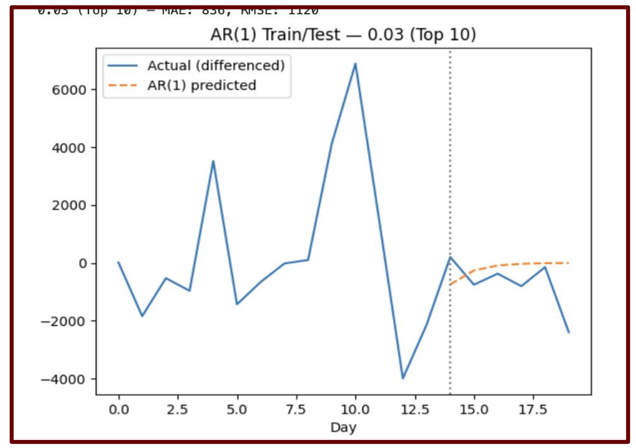
RESULTS

AR/MA Findings

The AR/MA pipeline processed 3,253 songs with a fixed ARMA(1,1) order. Class distribution: 2,374 non-hits (73%) vs 879 Top 10 hits (27%).



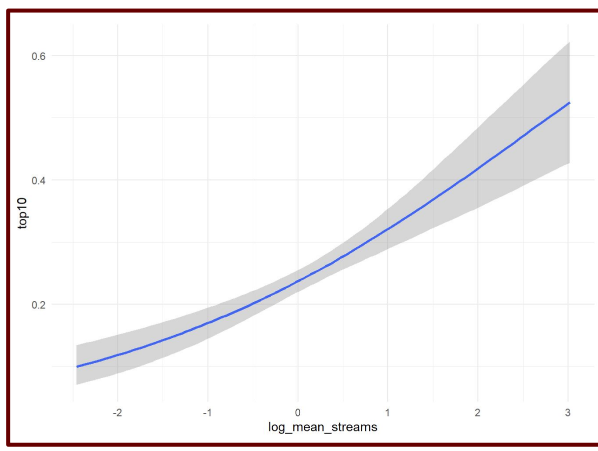


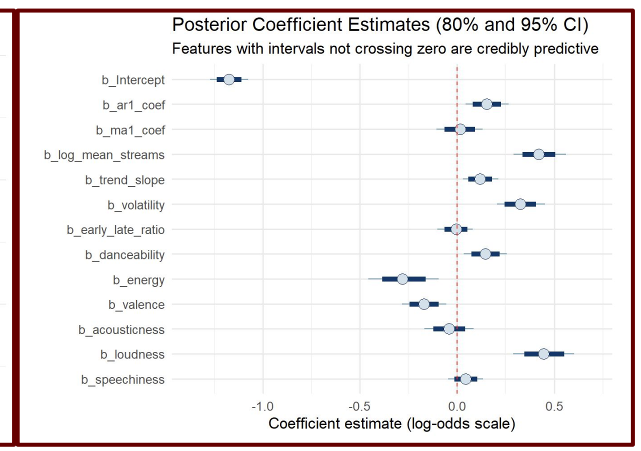


Bayesian Findings

The Bayesian model achieved a test accuracy of 72.7% on the held-out set (n=651). The confusion matrix reveals 438 true negatives and 34 true positives with only 14 false positives, indicating high precision.

```
## Regression Coefficients:
##                   Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept            -1.17      0.05    -1.27    -1.08 1.00     2864     2588
## ar1_coef              0.15      0.06     0.04     0.26 1.00     2865     2701
## ma1_coef              0.01      0.06    -0.11     0.13 1.00     2866     2626
## log_mean_streams      0.42      0.07     0.29     0.56 1.00     2750     2691
## trend_slope           0.12      0.05     0.03     0.21 1.00     2485     2549
## volatility            0.33      0.06     0.21     0.45 1.00     2923     2637
## early_late_ratio     -0.01      0.05    -0.10     0.08 1.00     2838     2891
## danceability          0.15      0.06     0.03     0.26 1.00     2405     2574
## energy               -0.28      0.09    -0.46    -0.09 1.00     2995     2990
## valence              -0.17      0.06    -0.28    -0.06 1.00     2887     2518
## acousticness         -0.04      0.06    -0.17     0.09 1.00     3039     2489
## loudness              0.45      0.08     0.29     0.60 1.00     2668     2756
## speechiness           0.04      0.05    -0.05     0.13 1.00     2872     2553
```





CONCLUSIONS

Trajectory-derived features add meaningful predictive value beyond audio attributes alone.

Key findings:

- Loudness is the strongest audio predictor so louder songs more likely to chart
- Log_mean_streams is the strongest predictor overall, as higher early stream volume strongly increases P(Top10) more than any audio feature
- Energy and valence are negatively associated which is surprising but consistent with recent trends toward mellower hits
- Ar1_coef and volatility are credibly positive, confirming trajectory in the first 3 weeks a valid signal

The main limitations are class imbalance (73% non-hits) which biases predictions toward the negative class, a low dataset join rate (~6%) limiting sample size.

NEXT STEPS

- Handle class imbalance via weighted likelihood or oversampling to improve recall on Top 10 hits
- Extend to multi-class prediction like Top 10 / Top 50 / Top 100 / Below
- Add artist-level random effects to account for popularity variation across artists

REFERENCES

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